

2026 MIDDLE SCHOOL STEM CAMPS

# PY-STEM

## INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

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# Magnetic Capsule Maze Cup

Steer a pill-camera without touching it

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**At a glance:** Steer a magnetic pill-camera through a body maze without touching it. About 80 min; 4 to 6 teams.

## MATERIALS

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- small neodymium magnets
- maze boards
- steel tokens or paper clips
- timers
- barrier sheets

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why does moving the magnet slowly help?
- How does distance change the pull you feel?
- Where is remote actuation used in real medicine?

SCORING (100 POINTS)

| SCORING CRITERION            | POINTS |
|------------------------------|--------|
| Fastest clean run            | 50     |
| Low penalties                | 20     |
| Magnetic-control explanation | 20     |
| Redesign                     | 10     |
| Total                        | 100    |

SOURCES Science Museum Group Magnetic Maze

# Oobleck Armor Arena

Armor from a liquid that hardens on impact

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**At a glance:** Design armor with a material that acts like a liquid until impact. About 80 min; 4 to 6 teams.

## MATERIALS

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- cornstarch
- water
- zip bags
- cardboard sleeves
- test weights
- spoons
- trays

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why does a fast press feel harder than a slow one?
- How little oobleck still protected the target?
- Where would shear-thickening armor be useful?

## SCORING (100 POINTS)

| SCORING CRITERION   | POINTS     |
|---------------------|------------|
| Protection result   | 40         |
| Material efficiency | 20         |
| Explanation         | 20         |
| Redesign            | 10         |
| Cleanup             | 10         |
| <b>Total</b>        | <b>100</b> |

**SOURCES** Science Buddies Oobleck; Exploratorium Ooze

# Cardboard Automata Arcade

One crank, one clever motion

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**At a glance:** Build a one-crank machine that performs a mini task. About 80 min; 4 to 6 teams.

## MATERIALS

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- cardboard
- skewers
- straws
- craft sticks
- tape
- hot glue station
- scissors

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- How did your cam shape change the motion?
- What caused jams, and how did you fix them?
- Where do cams show up in real machines?

## SCORING (100 POINTS)

| SCORING CRITERION            | POINTS     |
|------------------------------|------------|
| Reliable motion              | 30         |
| Cam or linkage complexity    | 20         |
| Task success                 | 20         |
| Explanation                  | 20         |
| Aesthetics and build quality | 10         |
| <b>Total</b>                 | <b>100</b> |

**SOURCES** Exploratorium Cardboard Automata

# Stethoscope Sprint and Recovery

Build a listening tool, measure like a scientist

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**At a glance:** Build a medical listening tool and use it like a sports scientist. About 80 min; 4 to 6 teams.

## MATERIALS

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- funnels
- flexible tubing
- tape
- balloons or membranes
- stopwatches
- sanitizing wipes

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- What change most improved your sound quality?
- Why does a good seal matter so much?
- What does a fast recovery suggest?



**SCORING (100 POINTS)**

| SCORING CRITERION    | POINTS |
|----------------------|--------|
| Sound quality        | 30     |
| Measurement accuracy | 25     |
| Explanation          | 20     |
| Redesign             | 15     |
| Data clarity         | 10     |
| Total                | 100    |

**SOURCES** Science Buddies Make a Stethoscope

# Reaction Time Combine

Prove which strategy cuts your delay

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**At a glance:** Train like a sports scientist and prove which strategy cuts delay. About 80 min; 4 to 6 teams.

## MATERIALS

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- meter sticks
- stopwatches
- cones
- score sheets
- clipboards

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why use the median instead of your fastest catch?
- Which strategy helped, and why might it?
- Where does reaction time matter in real life?

## SCORING (100 POINTS)

| SCORING CRITERION         | POINTS     |
|---------------------------|------------|
| Best median reaction time | 35         |
| Data quality              | 20         |
| Strategy improvement      | 20         |
| Explanation               | 15         |
| Teamwork                  | 10         |
| <b>Total</b>              | <b>100</b> |

**SOURCES** Science Buddies Reaction Time

# SONAR Slinky Showdown

Make invisible distance sensing visible

**At a glance:** Make invisible distance sensing visible with a slinky pulse. About 80 min; 4 to 6 teams.

## MATERIALS

- 4 to 6 metal slinkies
- floor tape
- station cards
- clipboards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- How is a slinky pulse like a sound wave?
- Why does timing an echo give distance?
- Where are ultrasonic sensors used?

## SCORING (100 POINTS)

| SCORING CRITERION     | POINTS     |
|-----------------------|------------|
| Model and predictions | 35         |
| Station performance   | 25         |
| Explanation           | 20         |
| Team communication    | 20         |
| <b>Total</b>          | <b>100</b> |

## SOURCES IIHS SONAR Slinky

# Pinhole Precision Challenge

The sharpest image, no lens

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**At a glance:** Build the sharpest camera image with no lens. About 80 min; 4 to 6 teams.

## MATERIALS

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- cardstock
- foil
- tape
- pushpins or paper clips
- white screen paper
- light source

## FACILITATION ARC

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- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why is the image upside down?
- What did shrinking the hole do, and why?
- How is this related to your eye?

## SCORING (100 POINTS)

| SCORING CRITERION   | POINTS     |
|---------------------|------------|
| Image clarity       | 35         |
| Concept explanation | 25         |
| Build quality       | 20         |
| Redesign            | 10         |
| Speed               | 10         |
| <b>Total</b>        | <b>100</b> |

**SOURCES** Exploratorium Personal Pinhole Theater; NASA JPL Pinhole Camera

# Low-Ropes Force Map Relay

Predict balance before the element proves you right

**At a glance:** Predict balance and stability before the ropes element proves you right. About 80 min; 4 to 6 teams.

## MATERIALS

- element cards
- clipboards
- center-of-mass templates
- pencils

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- When did your prediction match what you felt?
- How did lowering your center of mass help?
- Where else does balance physics matter?

## SCORING (100 POINTS)

| SCORING CRITERION   | POINTS     |
|---------------------|------------|
| Correct predictions | 35         |
| Force-map quality   | 25         |
| Debrief explanation | 20         |
| Teamwork reflection | 20         |
| <b>Total</b>        | <b>100</b> |

**SOURCES** Temple Ambler Outdoor Experiential Education; TeachEngineering balance references

# Hovercraft Hockey Hackathon

A puck that barely touches the floor

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**At a glance:** Build a puck that moves by barely touching the floor. About 80 min; 4 to 6 teams.

## MATERIALS

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- recycled CDs or plastic discs
- pop-top bottle caps
- balloons
- hot glue station
- targets
- floor tape

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why does the air cushion reduce friction?
- Why is more lift not always better?
- Where are air bearings used in real life?



## SCORING (100 POINTS)

| SCORING CRITERION        | POINTS     |
|--------------------------|------------|
| Glide performance        | 30         |
| Target competition score | 25         |
| Explanation              | 20         |
| Redesign                 | 15         |
| Build quality            | 10         |
| <b>Total</b>             | <b>100</b> |

**SOURCES** NASA JPL Hovering on a Cushion of Air; Science Buddies Hovercraft

# Spectra Sleuth Showdown

Two white lights, two hidden color fingerprints

**At a glance:** Can two white lights hide different color fingerprints? About 80 min; 4 to 6 teams.

## MATERIALS

- diffraction glasses or gratings
- LED sources
- incandescent source if available
- spectrum cards
- museum clue sheets

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- How can two white lights have different spectra?
- What does a line spectrum tell you?
- How do chemists make firework colors?

## SCORING (100 POINTS)

| SCORING CRITERION  | POINTS     |
|--------------------|------------|
| Correct matches    | 40         |
| Spectrum sketches  | 20         |
| Explanation        | 20         |
| Exhibit connection | 20         |
| <b>Total</b>       | <b>100</b> |

**SOURCES** Science History Institute Flash Bang Boom; Exploratorium Spectra

# BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

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**At a glance:** Retrieve the right book by using bin addresses, not a conveyor race. About 80 min; 4 to 6 teams.

## MATERIALS

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- bin cards
- address labels
- order deck
- route mat
- timers
- optional grabber tool

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why store by address instead of by subject?
- What made one route faster than another?
- Where else is this storage logic used?

## SCORING (100 POINTS)

| SCORING CRITERION                | POINTS     |
|----------------------------------|------------|
| Correct retrievals               | 35         |
| Low traffic conflicts            | 25         |
| Routing or algorithm explanation | 20         |
| Efficiency                       | 10         |
| Teamwork                         | 10         |
| <b>Total</b>                     | <b>100</b> |

**SOURCES** Temple Charles Library BookBot; Charles Library Circulating Collections

# Accessibility Ramp Rescue Lab

A portable ramp that meets real constraints

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**At a glance:** Design a portable ramp that works for a real user constraint. About 80 min; 4 to 6 teams.

## MATERIALS

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- foam board
- craft sticks
- dowels
- binder clips
- rulers
- weighted cart
- tape

## FACILITATION ARC

---

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

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- Why does a gentler slope need more length?
- Which criterion was hardest to meet?
- Why design for accessibility from the start?

## SCORING (100 POINTS)

| SCORING CRITERION           | POINTS     |
|-----------------------------|------------|
| Load held                   | 30         |
| Slope meets target          | 30         |
| Portability                 | 20         |
| Client criteria explanation | 20         |
| <b>Total</b>                | <b>100</b> |

**SOURCES** TeachEngineering Wheelchair Ramp Design

## BACKUP ACTIVITIES

# Ready alternates

# Pencil Pressure Safety Lab

Spread a force before it dents the target

**At a glance:** Spread a force before it dents the target. About 60 min; 4 to 6 teams.

## MATERIALS

- Pencils
- Foam
- Cardboard
- Weights
- Graph paper

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why does the same force do less damage over more area?
- Where is this used in safety gear?
- What made your spreader better?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS     |
|-------------------|------------|
| Pressure spread   | 40         |
| Explanation       | 25         |
| Data quality      | 20         |
| Redesign          | 15         |
| <b>Total</b>      | <b>100</b> |

**SOURCES** IIHS Pencil Pressure



# Domino Neuron Relay

Make a nerve signal survive a gap

**At a glance:** Make a nerve signal survive a gap. About 60 min; 4 to 6 teams.

## MATERIALS

- Dominoes
- Rulers
- Tape
- Index cards

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- What happened when the gap got too wide?
- How is a bridge piece like a synapse?
- Why does myelin speed real signals?

## SCORING (100 POINTS)

| SCORING CRITERION    | POINTS     |
|----------------------|------------|
| Transmission success | 35         |
| Gap challenge        | 25         |
| Explanation          | 20         |
| Redesign             | 20         |
| <b>Total</b>         | <b>100</b> |

**SOURCES** Exploratorium Domino Effect

# Pulley Rescue Relay

Lift the load with smarter force, not stronger arms

**At a glance:** Lift the load with smarter force, not stronger arms. About 60 min; 4 to 6 teams.

## MATERIALS

- Classroom pulleys
- String
- Weights
- Spring scales

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- Why did the movable pulley reduce effort?
- What did you trade for the easier lift?
- Where are pulley systems used in rescue work?

## SCORING (100 POINTS)

| SCORING CRITERION | POINTS     |
|-------------------|------------|
| Performance       | 35         |
| Setup accuracy    | 25         |
| Explanation       | 20         |
| Teamwork          | 20         |
| <b>Total</b>      | <b>100</b> |

**SOURCES** TryEngineering Pulleys and Force

# Barcode Checksum Rescue

Catch the fake barcode before the wrong part ships

**At a glance:** Catch the fake barcode before the warehouse ships the wrong part. About 60 min; 4 to 6 teams.

## MATERIALS

- Printed barcode cards
- Dry erase boards
- Timers

## FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

## DEBRIEF

- How does one extra digit catch a typo?
- Why do false alarms matter in a warehouse?
- Where do you see check digits in daily life?

## SCORING (100 POINTS)

| SCORING CRITERION  | POINTS     |
|--------------------|------------|
| Correct detections | 45         |
| Low false alarms   | 20         |
| Rule explanation   | 20         |
| Teamwork           | 15         |
| <b>Total</b>       | <b>100</b> |

**SOURCES** Computer science unplugged error-detection concepts